

08/04/2025



Aakash

Medical | IIT-JEE | Foundations

Corporate Office: Aakash Tower, 8, Pusa Road, New Delhi-110005, Ph.011-47623456

CODE-A



AIM - 720

(Advanced **INTENSIVE** Mastery for 720)

MM : 720

CST-I

Time : 180 Mins.

Complete Syllabus of Class XI & XII

Instructions:

- Duration of Test is 3 hrs.
- The Test consists of 180 questions. The maximum marks are 720.
- There are three parts in the question paper consisting of Physics, Chemistry, Biology which have 45 questions each in Physics & Chemistry and 90 questions in Biology.
- Each question carries +4 marks. For every wrong response, -1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
- Use blue/black ballpoint pen only to darken the appropriate circle.
- Mark should be dark and completely fill the circle.
- Dark only one circle for each entry.
- Dark the circle in the space provided only.
- Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

6 Aakashians scored a perfect 720/720 from Classroom Programs



720/720 | **Mridul M Anand**
3 Year Classroom



720/720 | **Ayush Naugraiya**
4 Year Classroom



720/720 | **Arghyadeep Dutta**
2 Year Classroom



720/720 | **Aryan Yadav**
1 Year Classroom



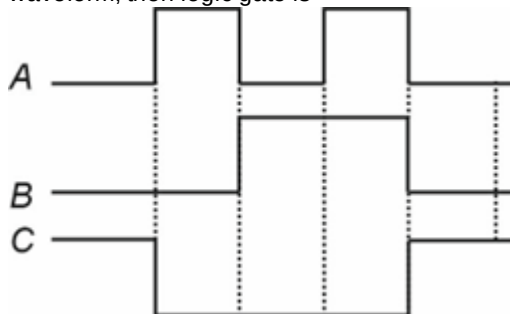
720/720 | **Palansha Agarwal**
2 Year Classroom



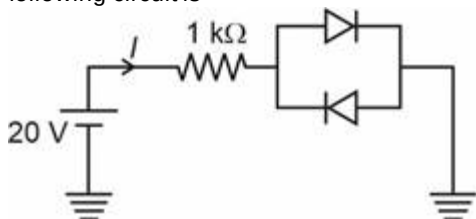
720/720 | **Iram Quazi**
1 Year Classroom

PHYSICS

1. For a logic gate A , B are input and C is output voltage waveform, then logic gate is



- (1) OR
(2) AND
(3) NOR
(4) NAND
2. Assuming the diodes to be ideal, the current I in the following circuit is



- (1) 10 mA
(2) 20 mA
(3) 10 A
(4) Zero
3. Light of wavelength 124 nm is incident on photo sensitive surface at a rate of 1.6×10^{-7} J/s. The number of photons falling per second on the surface will be
- (1) 10^9
(2) 10^{10}
(3) 10^{11}
(4) 10^{12}
4. In the reaction ${}^2_1\text{H} + {}^3_1\text{H} \rightarrow {}^4_2\text{He} + {}^1_0\text{n}$, if the binding energies of ${}^2_1\text{H}$, ${}^3_1\text{H}$ and ${}^4_2\text{He}$ are respectively a , b and c (in Mev), then the energy (in Mev) released in this reaction is
- (1) $a + b + c$
(2) $a + b - c$
(3) $c - a - b$
(4) $4c - 3b - 2a$

5. Length of a resonance tube is 120 cm. Number of resonances possible for wave having frequency 1320 Hz (speed of sound is 330 m s^{-1}) will be

- (1) 15
(2) 21
(3) 10
(4) 19

6. An equiconvex lens has refractive index μ and radius of curvature of each surface is R . It is cut into two symmetrical halves by a plane perpendicular to the principal axis. The power of one part is

- (1) $\frac{\mu-1}{2R}$
(2) $\frac{\mu-1}{R}$
(3) $\frac{2(\mu-1)}{R}$
(4) $\frac{2\mu-1}{R}$

7. Given below are two statements:

Statement A: About the centre of gravity of a body, the total gravitational torque on the body is non-zero.

Statement B: The centre of mas of symmetric bodies always coincides with their centre of gravity.

In the light of the above statements, choose the most appropriate answer from the options given below

- (1) Both statement A and statement B are correct
(2) Both statement A and statement B are incorrect
(3) Statement A is correct but statement B is incorrect
(4) Statement A is incorrect but statement B is correct

8. A voltmeter has a resistance R_0 and range V . The resistance which should be connected with it to increase its range to $3V$ will be

- (1) $2R_0$ in series
(2) $3R_0$ in series
(3) $2R_0$ in parallel
(4) $3R_0$ in parallel

9. If the ratio of the densities of two planets is $3 : 8$ and the ratio of their diameters is $2 : 3$, then ratio of their acceleration due to gravity (close to their surfaces) is

- (1) $1 : 1$
(2) $1 : 2$
(3) $3 : 1$
(4) $1 : 4$

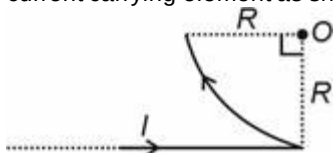
- 10. Assertion (A):** An induced emf must develop in conducting wire moving in a direction perpendicular to magnetic field.

Reason (R): EMF induced in a conducting wire of length

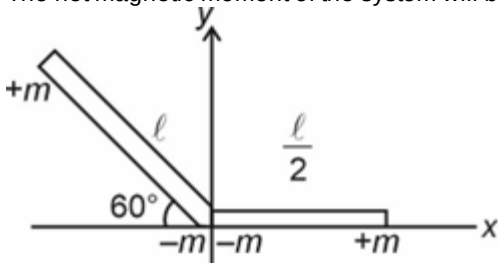
l moving with velocity $\vec{v}$ in magnetic field $\vec{B}$ is $(\vec{v} \times \vec{B}) \cdot \vec{l}$

In the light of the above statements, choose the correct answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 - (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
 - (3) (A) is false but (R) is true
 - (4) Both (A) and (R) are false
- 11.** The magnetic field at the centre O of circular part of the current carrying element as shown in the figure is



- (1) $\left(\frac{\mu_0 i}{4\pi R} - \frac{\mu_0 i}{8R}\right) \odot$ out of the plane of paper
 - (2) $\left(\frac{\mu_0 i}{2\pi R} - \frac{\mu_0 i}{2R}\right) \odot$ out of the plane of paper
 - (3) $\left(\frac{\mu_0 i}{4\pi R} - \frac{\mu_0 i}{4R}\right) \odot$ out of the plane of paper
 - (4) $\left(\frac{\mu_0 i}{4\pi R} - \frac{\mu_0 i}{2R}\right) \odot$ out of the plane of paper
- 12.** Two magnets are placed in x-y plane as shown in figure. The net magnetic moment of the system will be



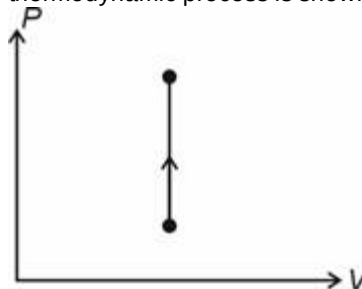
- (1) $\frac{\sqrt{3}ml}{4} \hat{i}$
 - (2) $\frac{\sqrt{3}ml}{2} \hat{j}$
 - (3) $\frac{ml}{4} \hat{i}$
 - (4) $\frac{ml}{4} \hat{j}$
- 13.** In an AC circuit, the voltage
- (1) Always lead the current
 - (2) Always lag the current
 - (3) Is always in phase of current
 - (4) May lead, lag or in phase with current

- 14.** Consider the following statements and tick the correct option.

Statement A: If a body is moving with constant speed but variable velocity then its path should be circular.

Statement B: If a body is moving with constant acceleration but variable velocity then its path should be straight line.

- (1) Statement A is correct while statement B is incorrect
 - (2) Statement A is incorrect while statement B is correct
 - (3) Both statement A and statement B are correct
 - (4) Both statement A and statement B are incorrect
- 15.** Consider the following statements:
- a. Linear and superficial expansion coefficients are not defined for liquids.
 - b. Due to increase in temperature, the time period of simple pendulum decreases.
 - c. At 4°C , the specific volume of water is minimum.
 - d. For heat transfer, two bodies must have same temperature
- Based on above information, pick correct option.
- (1) Only a and b are correct
 - (2) Only b and c are correct
 - (3) Only a, b and c are correct
 - (4) Only a and c are correct
- 16.** The P - V diagram for an ideal gas undergoing a thermodynamic process is shown in figure. The process is



- (1) Isothermal expansion
 - (2) Adiabatic compression
 - (3) Isochoric
 - (4) Isobaric expansion
- 17.** Consider the following statements:
- Statement A:** For an ideal gas, the change in internal energy of gas and work interaction of the gas are equal in magnitude in an adiabatic process.
- Statement B:** The internal energy does not change in an isothermal process for an ideal gas.
- Based upon above information, pick correct option.
- (1) Both statements A and B are correct
 - (2) Both statements A and B are incorrect
 - (3) Statement A is correct while B is incorrect
 - (4) Statement A is incorrect while B is correct

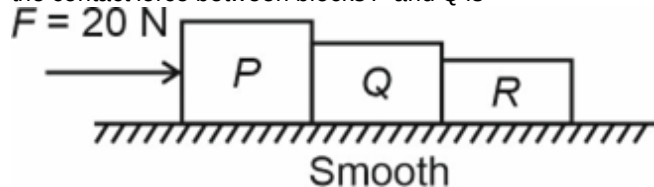
18. If a soap bubble contracts, the pressure inside the bubble
- Remains same
 - Decreases
 - Increases
 - Will become equal to atmospheric pressure
19. For a given gas at 1 atm pressure, rms speed of the molecules is v at 127°C . At 2 atm pressure, and at 227°C , the rms speed of molecules will be
- $\frac{\sqrt{5}}{4} v$
 - $\sqrt{\frac{5}{4}} v$
 - $\frac{1}{\sqrt{2}} v$
 - $\sqrt{\frac{3}{5}} v$
20. **Assertion (A):** Steel is more elastic than rubber.
Reason (R): Young's modulus is defined for solids only.
 In the light of the above statements, choose the correct answer from the options given below.
- Both (A) and (R) are true and (R) is the correct explanation of (A)
 - Both (A) and (R) are true but (R) is not the correct explanation of (A)
 - (A) is true but (R) is false
 - Both (A) and (R) are false
21. The kinetic energy and potential energy of a particle executing SHM are equal, when displacement in percentage terms of amplitude A is
- 50% of A
 - 70.7% of A
 - 60% of A
 - 25% of A
22. For the given stationary wave $y = 20\sin(4\pi x)\sin(20\pi t)$ where y and x are in m and t is in s. Match column I with column II.

Column-I	Column-II
a. Amplitude of component wave (in m)	(i) 5
b. Velocity of the component wave (in m s^{-1})	(ii) 0.125
c. Minimum separation between a node and an antinode (in m)	(iii) 0.5
d. Wavelength (in m) of the component wave	(iv) 10

- a(ii), b(i), c(iii), d(iv)
- a(iv), b(i), c(ii), d(iii)
- a(iv), b(ii), c(iii), d(i)
- a(i), b(ii), c(iii), d(iv)

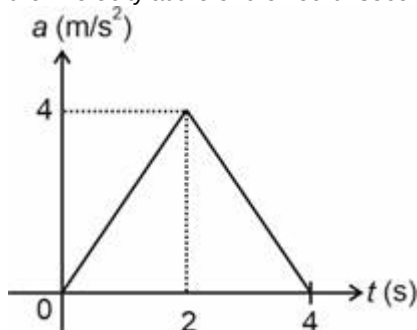
23. If U is the energy carried by EM wave travelling with speed c in free space, then momentum P of the EM wave is given by
- $P = Uc$
 - $P = \frac{U^2}{c}$
 - $P = \sqrt{Uc}$
 - $P = \frac{U}{c}$
24. Two coherent sources each emitting light of intensity I interfere in a medium at a point where phase difference between them is $\frac{\pi}{3}$, then resultant intensity at that point would be
- I
 - $4I$
 - $3I$
 - $\frac{I}{4}$
25. If velocity of a particle depends on time (t) as $v = p + qt + \frac{r}{s+t}$ then the dimensions of r would be $[p, q, r$ and s are positive constants]
- $[\text{T}^{-1}]$
 - $[\text{LT}^{-2}]$
 - $[\text{L}]$
 - $[\text{L}^{-1}\text{T}^{-1}]$
26. A particle is moving along x -axis. Its x -coordinate varies with time as, $x = 2t^2 + 4t - 6$. Here x is in metres and t is in seconds. The average velocity between the time interval $t = 0$ to $t = 2$ s is
- 2 m/s
 - 4 m/s
 - 6 m/s
 - 8 m/s

27. Three blocks P , Q and R of masses 5 kg, 4 kg and 1 kg respectively, are in contact on a frictionless surface as shown. If a force of 20 N is applied on the 5 kg block then the contact force between blocks P and Q is

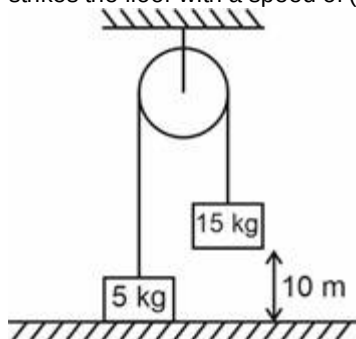


- 20 N
- 10 N
- 40 N
- 5 N

28. Acceleration-time graph of a particle moving in a straight line is as shown. If velocity of particle at time $t = 0$ is 4 m/s then velocity at the end of fourth second is



- (1) 4 m/s
(2) 12 m/s
(3) 8 m/s
(4) 6 m/s
29. Two masses, 15 kg and 5 kg are connected by an ideal string which is passing over an ideal pulley. If the mass 15 kg is released from the height 10 m above the floor, then it strikes the floor with a speed of ($g = 10 \text{ m s}^{-2}$)

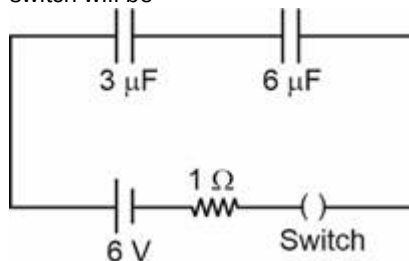


- (1) 10 m/s
(2) 30 m/s
(3) 50 m/s
(4) 12 m/s
30. **Assertion (A):** The work done by the conservative force depends only on the end points of a path.
Reason (R): $W_{\text{conservative}} = U_f - U_i$
In the light of the above statements, choose the correct answer from the options given below.
- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

31. An electron and a proton are placed at $1 \text{ \AA}$ distance apart, the electrostatic force between them is nearly

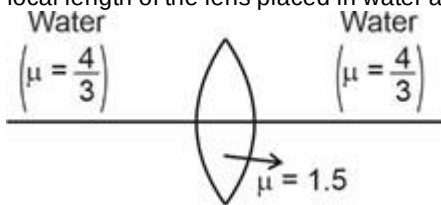
- (1) 144 nN
(2) 2.3 nN
(3) 23 nN
(4) 0.144 nN

32. Two uncharged capacitors are connected in series with a battery as shown in figure. After closing the switch, the net work done by the battery in a long time after closing the switch will be



- (1) $36 \mu\text{J}$
(2) $48 \mu\text{J}$
(3) $72 \mu\text{J}$
(4) $53 \mu\text{J}$
33. Consider the following statements.
Statement A: For a conducting wire, increase in drift velocity results in increase in current.
Statement B: For a conducting wire having constant potential difference across its ends, temperature increase results in decrease in drift velocity.
Mark the correct option
- (1) Statement A is correct while statement B is incorrect
(2) Statement A is incorrect while statement B is correct
(3) Both the statements are incorrect
(4) Both the statements are correct
34. Nine identical cells each of emf 2 V and internal resistance 1Ω are connected in series with load resistance of 11Ω and n cells out of these nine cells are reversed. If this combination of cells drive $\frac{1}{2} \text{ A}$ in a load resistance then the value of n is
- (1) 2
(2) 3
(3) 4
(4) Zero
35. In comparison to eye-piece of telescope, the objective has
- (1) Smaller focal length but larger aperture
(2) Smaller focal length and smaller aperture
(3) Larger focal length and larger aperture
(4) Larger focal length but smaller aperture

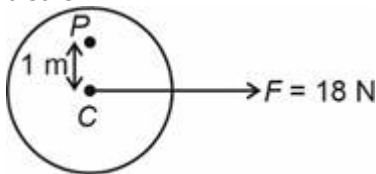
36. If radii of curvature of both convex surfaces is 30 cm, then focal length of the lens placed in water as shown is



- (1) 30 cm
(2) 120 cm
(3) 60 cm
(4) 90 cm
37. A thin uniform spherical shell has mass M and radius R with centre at O . Potential (V) due to this shell is given by

- (1) $V_{\text{centre}} = \frac{3}{2} V_{\text{surface}} = -\frac{3}{2} \frac{GM}{R}$
(2) $V_{\text{centre}} = \frac{2}{3} V_{\text{surface}} = -\frac{2}{3} \frac{GM}{R}$
(3) $V_{\text{centre}} = \frac{3}{2} V_{\text{surface}} = +\frac{3}{2} \frac{GM}{R}$
(4) $V_{\text{centre}} = V_{\text{surface}} = -\frac{GM}{R}$

38. A uniform disc of mass 10 kg and radius 2 m is hinged about point P . A force 18 N acts in the plane of the disc at its centre as shown below. Initial angular acceleration of the disc is



- (1) $\frac{2}{3} \text{ rad/s}^2$
(2) $\frac{2}{5} \text{ rad/s}^2$
(3) $\frac{3}{5} \text{ rad/s}^2$
(4) $\frac{4}{3} \text{ rad/s}^2$
39. A current carrying infinite wire is passing through the centre of a ring of radius R and perpendicular to plane of the ring. The magnetic flux passing through the ring will be

- (1) $\frac{\mu_0 IR}{4}$
(2) $\frac{\mu_0 IR}{2}$
(3) $\mu_0 IR$
(4) Zero

40. In an R-L-C series circuit, the voltage and current are represented by $V = 100\sin(100\pi t)$ V and $I = 20\sin\left(100\pi t + \frac{\pi}{3}\right)$ A. The value of resistance in the circuit is

- (1) 5 Ω
(2) 3 Ω
(3) $\frac{5}{3} \Omega$
(4) $\frac{5}{2} \Omega$

41. If wavelength of electron in third Bohr orbit is 10^{-8} m, then the circumference of the orbit is

- (1) 9×10^{-8} m
(2) 4×10^{-8} m
(3) 10^{-8} m
(4) 3×10^{-8} m

42. Suppose a pure Si crystal has 4×10^{26} atoms m^{-3} . It is doped by 1 ppm concentration of pentavalent P . Calculate the number density of electrons and holes. (Given that $n_i = 1.4 \times 10^{16} \text{ m}^{-3}$)

- (1) $4 \times 10^{20} \text{ m}^{-3}, 4.9 \times 10^{10} \text{ m}^{-3}$
(2) $4 \times 10^{26} \text{ m}^{-3}, 4 \times 10^{20} \text{ m}^{-3}$
(3) $4 \times 10^{20} \text{ m}^{-3}, 4.9 \times 10^{11} \text{ m}^{-3}$
(4) $4 \times 10^{26} \text{ m}^{-3}, 4.9 \times 10^{10} \text{ m}^{-3}$

43. A flat circular ring of radius 4.5 cm is placed gently over the surface of water. If surface tension of water is 0.07 N m^{-1} , then excess force required to pull it away from surface is

- (1) 19.8 mN
(2) 39.6 mN
(3) 3.96 mN
(4) 1.98 mN

44. If the radius of star is R and it acts as a blackbody, then temperature of star if the rate of radiation is Q will be

(1) $\left(\frac{Q}{4\pi R^2 \sigma}\right)^{-\frac{1}{4}}$

(2) $\left(\frac{4\pi R^2 Q}{\sigma}\right)^{\frac{1}{4}}$

(3) $\left(\frac{Q}{4\pi R^2 \sigma}\right)^{-\frac{1}{2}}$

(4) $\left(\frac{Q}{4\pi R^2 \sigma}\right)^{\frac{1}{4}}$

45. A beam of light of wavelength $5000 \text{ \AA}$ from a distance source falls on a single slit 2 mm wide and resulting diffraction pattern is observed on a screen 2 m away. Width of central bright fringe will be

(1) 0.5 mm

(2) 2 mm

(3) 5 mm

(4) 1 mm

CHEMISTRY

46. In which of the following pairs of molecules/ions, the central atoms have sp^2 hybridisation?

(1) NO_2^- and NH_3

(2) BF_3 and NO_2^-

(3) BF_3 and NH_2^-

(4) NH_2^- and H_2O

47. The correct increasing order of ionic character is

(1) $\text{LiCl} < \text{NaCl} < \text{KCl}$

(2) $\text{KCl} < \text{LiCl} < \text{NaCl}$

(3) $\text{NaCl} < \text{LiCl} < \text{KCl}$

(4) $\text{KCl} < \text{NaCl} < \text{LiCl}$

48. A solution of sucrose has been prepared by dissolving 34.2 g of sucrose in 1000 g of water. The freezing point of the solution obtained will be (K_f for water = $1.86 \text{ K kg mol}^{-1}$)

(1) -0.186°C

(2) $+0.372^\circ\text{C}$

(3) -0.372°C

(4) $+0.186^\circ\text{C}$

49. Consider the following statements.

Statement I: Solubility of gases in liquids increases with increase in temperature.

Statement II: For a solution of volatile liquids, the partial vapour pressure of each volatile component of the solution is directly proportional to its mole fraction in solution. In the light of above statements, choose the most appropriate answer from the options given below

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is incorrect but statement II is correct

(4) Statement I is correct but statement II is incorrect

50. Consider the following statements.

a. BeH_2 and BCl_3 have incomplete octet of central atoms.

b. N_2O is an odd electron molecule.

c. PF_5 and SF_6 have expanded octet of central atom.

The correct statements are

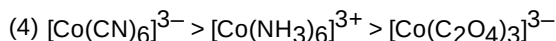
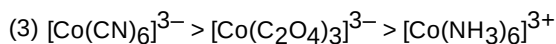
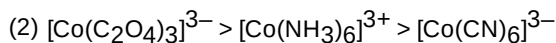
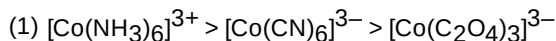
(1) a and b only

(2) b and c only

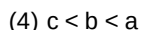
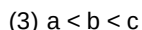
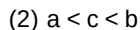
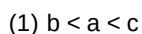
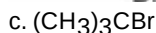
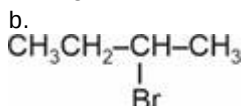
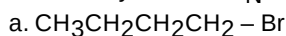
(3) a and c only

(4) a, b and c

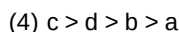
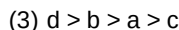
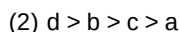
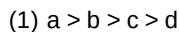
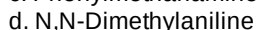
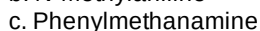
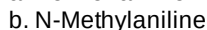
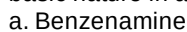
51. The correct decreasing order for the wavelength of absorption for the complexes of Co^{3+} is



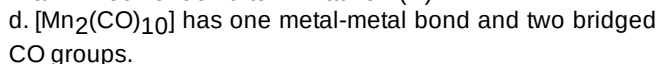
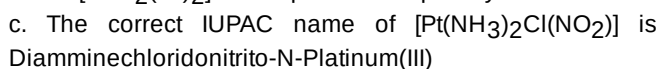
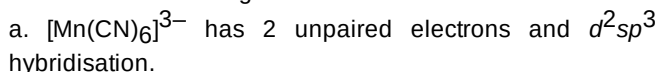
52. Arrange the following compounds in their increasing order of reactivity towards $\text{S}_{\text{N}}1$ reaction



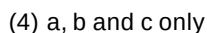
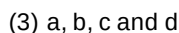
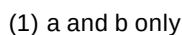
53. Arrange the following amines in their decreasing order of basic nature in aqueous solution



54. Consider the following statements



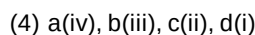
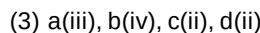
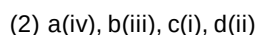
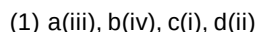
The correct statements are



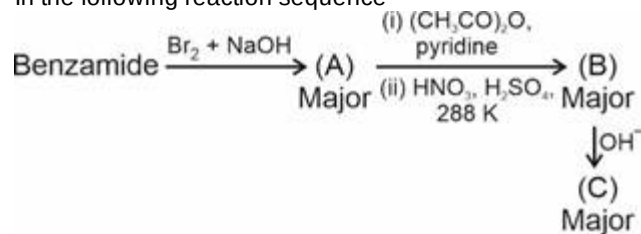
55. Match column-I with column-II

	Column-I (Reaction)		Column-II (Name of reaction)
a.		(i)	Finkelstein
b.		(ii)	Swarts
c.		(iii)	Sandmeyer's
d.		(iv)	Gatterman

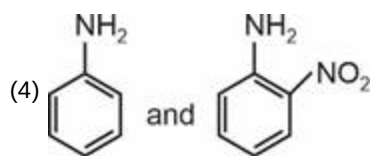
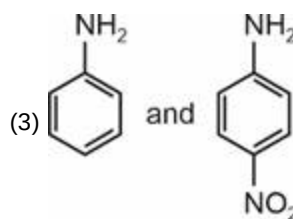
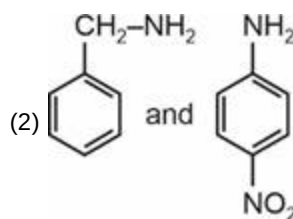
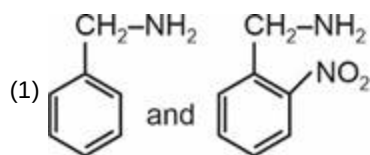
The correct match is



56. In the following reaction sequence



The products (A) and (C) respectively are

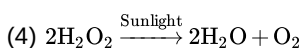
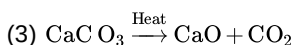
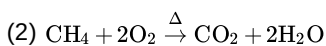
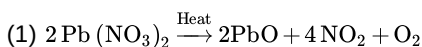


57. Match the column-I containing some inorganic compounds with column-II containing their chemical formula.

	Column-I		Column-II
a.	Kernite	(i)	$\text{Na}_2\text{B}_4\text{O}_7 \times 10\text{H}_2\text{O}$
b.	Borax	(ii)	$\text{Na}_2\text{B}_4\text{O}_7 \times 4\text{H}_2\text{O}$
c.	Bauxite	(iii)	$\text{Al}_2\text{O}_3 \times 2\text{H}_2\text{O}$
d.	Cryolite	(iv)	Na_3AlF_6

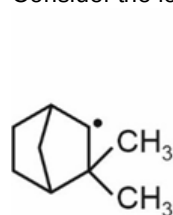
Choose the correct answer from the options given below.

- (1) a(i), b(iii), c(iv), d(ii)
 (2) a(ii), b(i), c(iv), d(iii)
 (3) a(i), b(ii), c(iv), d(iii)
 (4) a(ii), b(i), c(iii), d(iv)
58. Choose the incorrect statement
- (1) n-Pentane and isopentane are chain isomers
 (2) 2,2-Dimethylpropane is also known as neopentane
 (3) Propanone and propanal are functional group isomers
 (4) Ethoxyethane and ethoxypropane are metamers
59. Which of the given reactions is an example of disproportionation?

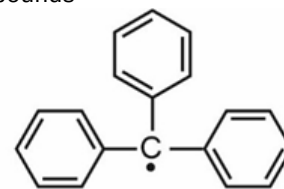


60. Consider the given redox reaction in acidic medium
 $\text{MnO}_4^- + \text{C}_2\text{O}_4^{2-} + \text{H}^+ \rightarrow \text{Mn}^{2+} + \text{CO}_2 + \text{H}_2\text{O}$
 The number of moles of KMnO_4 required to oxidise 0.5 mol of oxalate ions is
- (1) 0.1 mol
 (2) 0.2 mol
 (3) 0.5 mol
 (4) 1 mol

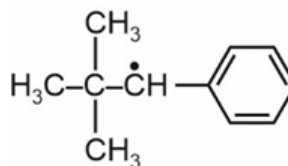
61. Consider the following compounds



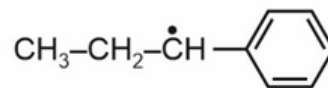
(I)



(II)



(III)



(IV)

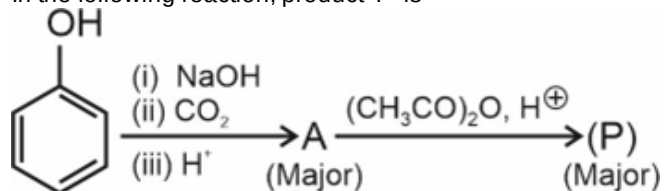
Hyperconjugation does not occur in

- (1) (I) only
 (2) (I), (II) and (III) only
 (3) (IV) only
 (4) (I), (II), (III) and (IV)
62. Select the incorrect statement.
- (1) Zinc reacts with concentrated nitric acid to give nitrous oxide.
 (2) Chromium does not dissolve in concentrated nitric acid
 (3) Concentrated nitric acid oxidises iodine to iodic acid
 Phosphine is prepared by heating white phosphorus with
 (4) concentrated NaOH solution in an inert atmosphere of CO_2

63. Chemical formula of mustard gas is

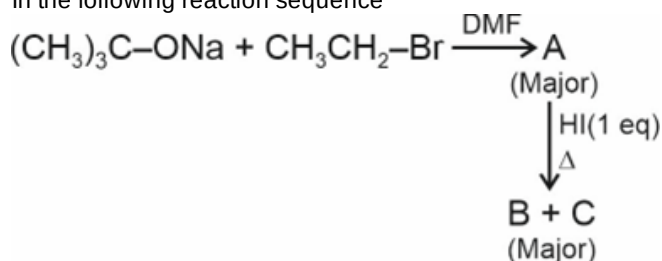
- (1) COCl_2
 (2) $\text{ClCH}_2\text{CH}_2\text{SCH}_2\text{CH}_2\text{Cl}$
 (3) CCl_3NO_2
 (4) CHCl_3

64. In the following reaction, product 'P' is



- (1) Salicylaldehyde
 (2) Benzoquinone
 (3) Salicylic acid
 (4) Aspirin

65. In the following reaction sequence



The product B and C is

- (1) $(\text{CH}_3)_3\text{C}-\text{I}$, $\text{CH}_3\text{CH}_2\text{OH}$
- (2) $\text{CH}_3\text{CH}_2-\text{I}$, $(\text{CH}_3)_3\text{C}-\text{OH}$
- (3) $\text{CH}_3-\text{CH}_2-\text{I}$, $\text{CH}_3-\text{CH}_2-\text{OH}$
- (4) $(\text{CH}_3)_2\text{CH}-\text{OH}$, $\text{CH}_3\text{CH}_2-\text{I}$

66. Which of the following will not be formed on complete hydrolysis of XeF_4 ?

- (1) Xe
- (2) XeO_3
- (3) F_2
- (4) HF

67. Match the column-I with column-II

Column-I Compound	Column-II pK_a
a. o-Nitrophenol (i)	10.2
b. p-Nitrophenol (ii)	10.1
c. m-Cresol (iii)	7.1
d. p-Cresol (iv)	7.2

Choose the correct answer from the options given below.

- (1) a(iii), b(iv), c(ii), d(i)
- (2) a(iii), b(iv), c(i), d(ii)
- (3) a(iv), b(iii), c(ii), d(i)
- (4) a(iv), b(iii), c(i), d(ii)

68. If solubility of $\text{Al}_2(\text{SO}_4)_3$ is 's' mol L^{-1} , then the solubility product (K_{sp}) is

- (1) $72s^5$
- (2) $6s^2$
- (3) $54s^4$
- (4) $108s^5$

69. Which among the following is the correct order of metallic character of the given elements?

- (1) $\text{K} > \text{Al} > \text{Mg} > \text{B}$
- (2) $\text{B} > \text{Al} > \text{Mg} > \text{K}$
- (3) $\text{K} > \text{Mg} > \text{Al} > \text{B}$
- (4) $\text{Mg} > \text{K} > \text{Al} > \text{B}$

70. On electrolysis of H_2O , how many faradays are required to produce 16 g of oxygen gas at anode?

- (1) 2
- (2) 4
- (3) 3
- (4) 6

71. The dissociation constant of an acid HA is 10^{-6} . The dissociation constant of its conjugate base at 25°C is

- (1) 10^{-3}
- (2) 10^{-6}
- (3) 10^{-8}
- (4) 10^{-9}

72. The molar conductivity of 0.2 M NaCl aqueous solution having conductivity of $1.6 \times 10^{-3} \text{ S cm}^{-1}$ is

- (1) $2 \text{ S cm}^2 \text{ mol}^{-1}$
- (2) $4 \text{ S cm}^2 \text{ mol}^{-1}$
- (3) $8 \text{ S cm}^2 \text{ mol}^{-1}$
- (4) $16 \text{ S cm}^2 \text{ mol}^{-1}$

73. Given below are two statements,

Statement I: Electron gain enthalpy of bromine is more negative than fluorine.

Statement II: Silicon is more electronegative than carbon as per Pauling scale.

In the light of above statements, choose the correct option.

- (1) Statement I is correct but statement II is incorrect
- (2) Statement I is incorrect but statement II is correct
- (3) Both statement I and statement II are incorrect
- (4) Both statement I and statement II are correct

74. Given below are two statements

Statement I: Molecularity of a reaction cannot be zero.

Statement II: Order of reaction is experimental quantity and can be fraction or zero.

In the light of above statements, choose the most appropriate answer from the options given below

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is incorrect but statement II is correct
- (4) Statement I is correct but statement II is incorrect

75. Reaction of acetone with Grignard reagent followed by hydrolysis majorly gives

- (1) Tertiary alcohol
- (2) Carboxylic acid
- (3) Secondary alcohol
- (4) Alkane

76. Which among the following salt solution will give rotten egg like smell on addition of dilute H_2SO_4 ?

- (1) $\text{NaCl}(\text{aq})$
- (2) $\text{Na}_2\text{S}(\text{aq})$
- (3) $\text{KNO}_2(\text{aq})$
- (4) $\text{Na}_2\text{CO}_3(\text{aq})$

77. Match group of cation in column-I with their group reagent of qualitative analysis in column-II

	Column-I		Column-II
a.	Group-I	(i)	$\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$
b.	Group-II	(ii)	dilute HCl
c.	Group-III	(iii)	$\text{H}_2\text{S} + \text{dilute HCl}$
d.	Group-IV	(iv)	$\text{H}_2\text{S} + \text{NH}_4\text{OH}$

Correct match among the following is

- (1) a(iii), b(ii), c(iv), d(i)
- (2) a(iii), b(ii), c(i), d(iv)
- (3) a(ii), b(iii), c(iv), d(i)
- (4) a(ii), b(iii), c(i), d(iv)

78. If half life of a first order reaction is 24 min, then time required to consume 93.75% reactant will be

- (1) 48 min
- (2) 72 min
- (3) 120 min
- (4) 96 min

79. Correct decreasing order of pK_a for given acids is

- a. CH_3COOH
- b. HCOOH
- c. $\text{C}_6\text{H}_5\text{COOH}$

- (1) $a > b > c$
- (2) $b > c > a$
- (3) $a > c > b$
- (4) $c > b > a$

80. Match the following aspects with the respective metal of 3d series.

	List-I (Aspects)		List-III (Metal)
a.	The metal with highest enthalpy of atomisation	(i)	Manganese
b.	The metal which in +1 oxidation state undergoes disproportionation in aqueous solution	(ii)	Vanadium
c.	The metal with highest spin only magnetic moment in +2 oxidation state	(iii)	Copper
d.	The metal which does not exhibit variable oxidation states	(iv)	Scandium

Select the correct option

- (1) a(ii), b(iii), c(i), d(iv)
- (2) a(ii), b(iv), c(iii), d(i)
- (3) a(iii), b(ii), c(i), d(iv)
- (4) a(iii), b(i), c(ii), d(iv)

81. Which one of the followings has maximum number of molecules among the following?

- (1) 18 mL of water
- (2) 0.018 g of water
- (3) 10^{-2} mol of water
- (4) 0.00224 L of water vapours at 1 atm and 273 K

82. The correct relationship between C_P molar heat capacity at constant pressure and C_V molar heat capacity at constant volume for 2 moles of an ideal gas is

- (1) $C_P + C_V = 2R$
- (2) $C_P - C_V = R$
- (3) $C_P - C_V = 2R$
- (4) $C_P - C_V = \frac{R}{2}$

83. Given below are two statements

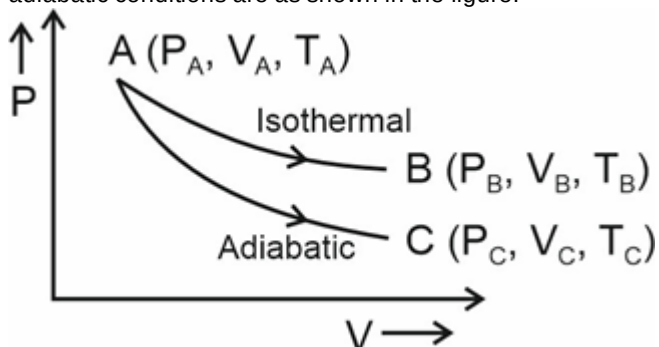
Statement I: Due to lanthanoid contraction, Zr and Hf have almost identical radii.

Statement II: Both Ce^{4+} and Yb^{2+} are diamagnetic in nature.

In the light of above statements, choose the most appropriate answer from the options given below

- (1) Both statement I and statement II are incorrect
- (2) Both statement I and statement II are correct
- (3) Statement I is incorrect but statement II is correct
- (4) Statement I is correct but statement II is incorrect

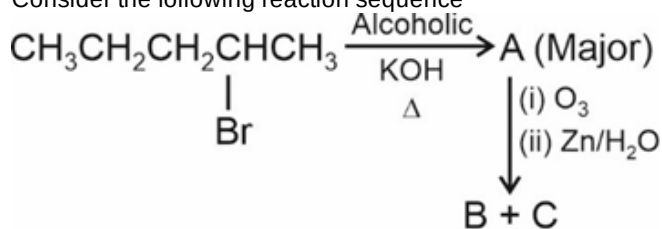
84. Reversible expansion of an ideal gas under isothermal and adiabatic conditions are as shown in the figure.



- a. $\Delta U_{\text{isothermal}} < \Delta U_{\text{adiabatic}}$ (Magnitude only)
 b. $T_C > T_B$
 c. $T_B = T_A$
 d. $w_{\text{isothermal}} < w_{\text{adiabatic}}$ (By system)
 Identify the incorrect statements.

- (1) a, b and c only
 (2) b and c only
 (3) b and d only
 (4) a, b and d only

85. Consider the following reaction sequence



Major product B and C are

- (1) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$ and HCHO
 (2) $\text{CH}_3\text{CH}_2\text{CHO}$ and CH_3CHO
 (3) $\text{CH}_3\text{CH}_2\text{COOH}$ and CH_3COOH
 (4) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ and $\text{CH}_3\text{CH}_2\text{OH}$

86. If a particle of mass 5 mg is moving with a velocity of 2000 m s^{-1} then the wavelength associated with the moving particle will be ($h = 6.6 \times 10^{-34} \text{ Js}$)

- (1) $6.6 \times 10^{-35} \text{ m}$
 (2) $6.6 \times 10^{-32} \text{ m}$
 (3) $6.6 \times 10^{-30} \text{ m}$
 (4) $6.6 \times 10^{-28} \text{ m}$

87. Match the column-I with column-II

Column-I Vitamin	Column-II Deficiency disease
a. Pyridoxine	(i) Cheilosis
b. Thiamine	(ii) Scurvy
c. Riboflavin	(iii) Convulsions
d. Ascorbic acid	(iv) Beri beri

Choose the correct answer from the options given below

- (1) a(iii), b(iv), c(ii), d(i)
 (2) a(ii), b(iii), c(iv), d(i)
 (3) a(iv), b(iii), c(i), d(ii)
 (4) a(iii), b(iv), c(i), d(ii)

88. Given below are the two statements

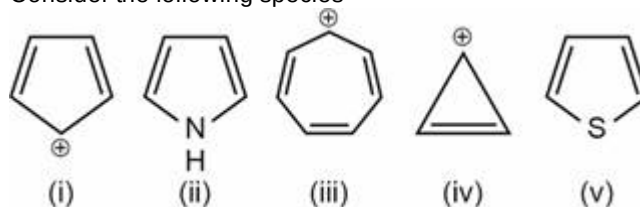
Statement I: Amylopectin is insoluble in water and constitutes about 80-85% of starch

Statement II: Chemically amylopectin is a long unbranched chain with 200-1000 α -D-(+)-glucose units held together by C1-C4 glycosidic linkage.

In the light of above statements, choose the most appropriate answer from the options given below

- (1) Statement I is correct but statement II is incorrect
 (2) Statement I is incorrect but statement II is correct
 (3) Both statement I and statement II are correct
 (4) Both statement I and statement II are incorrect

89. Consider the following species



The total number of aromatic species are

- (1) 2
 (2) 4
 (3) 5
 (4) 3

90. Correct order of energy of the given orbitals for multielectronic species is

- (1) $5s > 4s > 4d > 3d$
 (2) $4d > 5s > 4s > 3d$
 (3) $5s > 4d > 4s > 3d$
 (4) $4d > 5s > 3d > 4s$

BIOLOGY

91. Match column I with column II and select the **correct** option.

Column -I (Biological name)	Column-II (Taxonomic categories)
a. <i>Musca domestica</i>	(i) Poales
b. <i>Mangifera indica</i>	(ii) Polymoniales
c. <i>Solanum melongena</i>	(iii) Diptera
d. <i>Triticum aestivum</i>	(iv) Sapindales

(1) a(iii); b(iv); c(ii); d(i)

(2) a(iii); b(iv); c(i); d(ii)

(3) a(iv); b(iii); c(ii); d(i)

(4) a(iv); b(iii); c(i); d(ii)

92. Five kingdom classification was proposed by

- (1) R. H. Whittaker
- (2) Linnaeus
- (3) Aristotle
- (4) Carl Woese

93. A prokaryotic organism 'X' that completely lack a cell wall and is smallest living cell known, can survive without oxygen. This organism 'X' will most likely to be

- (1) *Nostoc*
- (2) Methanogens
- (3) *Mycoplasma*
- (4) Halophiles

94. Select the **mismatched** pair.

- (1) Stilt roots – Maize
- (2) Respiratory roots – *Rhizophora*
- (3) Storage roots – Zaminkand
- (4) Prop roots – Banyan tree

95. In which of the following plants fruits are formed without fertilisation?

- (1) Apple
- (2) Mango
- (3) Grapes
- (4) Coconut

96. Read the following statements (a-d).

- a. Cell wall is made up of peptidoglycan
 - b. Cytoplasmic ribosomes are made up of 60S and 40S subunits
 - c. Cyanobacteria oxidise inorganic substances and use the released energy for ATP production hence are chemoautotrophs.
 - d. Some of organisms are autotrophic but vast majority are heterotrophic
 - e. Respiratory enzymes are found associated with plasma membrane
- How many of the above features are seen in the members of kingdom Monera?

(1) Four

(2) Three

(3) Two

(4) Five

97. Read the following statements (a-d).

- a. In lily, the stamen is attached to the perianth and this condition is said to be epiphyllous
 - b. In brinjal, the stamen is attached to the petal, and this condition is called epipetalous
 - c. In Australian acacia, the petioles modify to form flat, green-coloured leaf-like structure which perform the function of photosynthesis
 - d. In cymose inflorescence, young flowers are present towards the apex and the older flowers are present at the base
- In the light of above statements select the **correct** ones from the option given below.

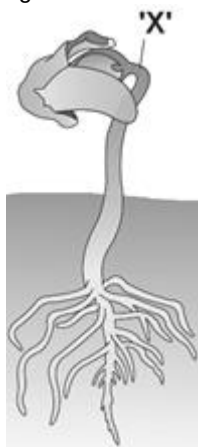
(1) Only a, c and d

(2) Only a and b

(3) Only a, b and c

(4) Only b and d

98. Select the **correct** option for the label 'X' in the following figure.

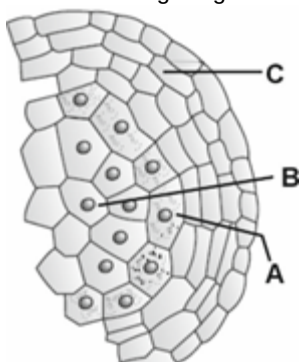


- (1) Cotyledon
(2) Seed coat
(3) Hypocotyl
(4) Epicotyl hook
99. Select the **correct** statement w.r.t. cotton
- (1) It lacks plasticity
(2) Its growth is stimulated by dormin
(3) Shapes of leaves that are produced in air are different than those produced in water
(4) Its abscission is accelerated by ethylene

100. The basal part of the ovule is represented by

- (1) Funicle
(2) Hilum
(3) Micropyle
(4) Chalaza

101. Observe the figure given below.



Choose the **correct** labellings for 'A', 'B' and 'C' from the following options.

- (1) A-Microspore mother cells, B- Tapetum, C-Epidermis
(2) A-Microspore mother cells, B- Endothecium, C-Epidermis
(3) A-Tapetum, B- Microspore mother cells, C-Endothecium
(4) A-Tapetum, B- Microspore mother cells, C-Epidermis

102. Following are given two statements, labelled as Assertion (A) and Reason(R).

Assertion(A): Distribution of bryophytes and pteridophytes is limited.

Reason(R): Pollination occurs by water in bryophytes and pteridophytes.

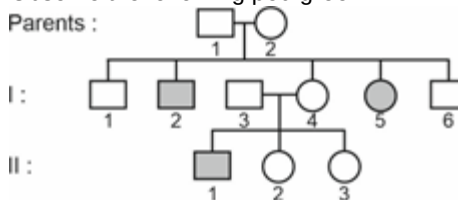
In the light of the above statements, identify the **correct** option from the following.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) (R) is true but (A) is false
103. Which of the following is a recessive trait of the pea plant selected by Mendel for his studies?
- (1) Axial floral position
(2) Violet colour of flower
(3) Yellow seed
(4) Constricted pod

104. Which of the following depicts the **correct** genotypic ratio in the F₂ generation if a cross is conducted between a tall pea plant and a dwarf pea plant?

- (1) 1 : 1
(2) 9 : 3 : 3 : 1
(3) 1 : 2 : 1
(4) 3 : 1

105. Observe the following pedigree



Based on the above pedigree, certain inferences were proposed.

- a. All the affected individuals are haemophilic.
b. This pedigree exemplifies the inheritance pattern of myotonic dystrophy.
c. Parents are homozygous recessive for the trait under study.
d. The affected individuals are homozygous recessive for the trait.
e. If I-5 marries with an unaffected homozygous male, then all progenies will be unaffected
Select the **correct** one(s).

- (1) a and b only
(2) b, c and d only
(3) a, b and c only
(4) d and e

- 106.** A closer examination of human body and a sample of Earth's crust reveals that the relative abundance of
- (1) Oxygen is higher in human body than in Earth's crust
 - (2) Carbon is higher in Earth's crust than in human body
 - (3) Nitrogen is same in human body as well as in sample of Earth's crust
 - (4) Hydrogen is higher in Earth's crust than in human body

- 107.** Complete the analogy by selecting the **correct** option.
Acid-soluble fraction : Phosphate :: Acid-insoluble fraction :

- (1) Glycine
- (2) Glucose
- (3) Guanine
- (4) Glycogen

- 108.** If a healthy person is running fast, then
- a. Every 100 mL of oxygenated blood will deliver more than 5 mL of oxygen to tissues.
 - b. Duration of inspiration will reduce.
 - c. Cardiac output will increase.
- Choose the **correct** option.

- (1) Only a and b are correct
- (2) Only c is correct
- (3) Only b and c are incorrect
- (4) a, b and c are correct

- 109.** The type of epithelium that is present in air sacs of lungs possesses _____ cells. Choose the option that **correctly** fills the blank.

- (1) Tall and slender
- (2) Cube-like
- (3) Ciliated
- (4) Flattened

- 110.** If we completely hydrolyse cytidylic acid, then all of the following will be produced, **except**

- (1) A hexose sugar
- (2) A pyrimidine derivative
- (3) Phosphoric acid
- (4) Ribose

- 111.** Presence of how many of the features mentioned in the box below is/are not considered as fundamental characteristic(s) of chordates?

Notochord, Dorsal heart, Dorsal solid CNS, Pharyngeal gill-slits, Post-anal tail

Select the **correct** option.

- (1) One
- (2) Two
- (3) Three
- (4) Four

- 112.** Read the statements A and B and choose the most appropriate option.

Statement A: Animals never accumulate ammonia, urea, CO₂, Na⁺, K⁺, phosphate, sulphate, etc., either by metabolic activities or by excess ingestion.

Statement B: In bony fishes, ammonia is generally excreted by diffusion through gill surfaces as ammonium ions.

- (1) Both the statements A and B are true
- (2) Only statement A is true
- (3) Only statement B is true
- (4) Both the statements A and B are false

- 113.** Dinosaurs were dominant on Earth during which of the following geological times?

- (1) Paleozoic era
- (2) Mesozoic era
- (3) Carboniferous period
- (4) Permian period

- 114.** In 1953, S. L. Miller created electric discharge in a closed flask that contained all of the following at 800°C, **except**

- (1) Hydrogen
- (2) Oxygen
- (3) Ammonia
- (4) Methane

- 115.** Read the statements given below w.r.t. evolution.

(A) The first non-cellular forms of life could have originated 3 bya.

(B) The first cellular forms of life could have originated about 2000 bya.

(C) Louis Pasteur by careful experimentation demonstrated that life came out of decaying and rotting matter.

Which of the above statements is/are not **incorrect**?

- (1) Only A
- (2) Only B
- (3) A and B only
- (4) B and C only

116. Mark the given statements as **True(T)** or **False(F)** and select the **correct** option.

- Deletions and insertions of DNA cause substitution mutation.
- UV rays can act as mutagens.
- Cancer cells commonly show chromosomal abnormalities

a b c

- T T F
- T F T
- F T T
- F F T

- (1)
- (2)
- (3)
- (4)

117. Which of following elements of phloem is absent in most of the monocotyledons?

- Sieve tubes
- Companion cells
- Phloem parenchyma
- Phloem fibres

118. Read the given **Assertion (A)** and **Reason (R)** and choose the **correct** option.

Assertion(A): In roots, pericycle and hypodermis provide mechanical support.

Reason(R): In roots, pericycle is made up of sclerenchyma while hypodermis is made up of collenchymatous cells.

- (A) is true but (R) is false
- Both (A) and (R) are true but (R) is not the correct explanation of (A)
- Both (A) and (R) are false
- Both (A) and (R) are true and (R) is the correct explanation of (A)

119. In eukaryotes, which rRNA is **not** transcribed by RNA polymerase I among those given below?

- 5.8 S rRNA
- 28 S rRNA
- 5 S rRNA
- 18 S rRNA

120. Read the given statements and choose the **correct** option.

Statement A: DNA fingerprinting is a technique to find out variations in individuals of a population at DNA level.

Statement B: In *lac* operon, the repressor is synthesised constitutively from the *i* gene.

- Both statements A and B are correct
- Both statements A and B are incorrect
- Only statement A is correct
- Only statement B is correct

121. Select the **correctly** matched pair.

- | | |
|-----------------------------------|---|
| (1) Friedrich Meischer | – Discovery of nuclein in 1896 |
| (2) DNA gyrase | – Topoisomerase activity in prokaryotes |
| (3) Ribonucleic acid | – Mutate at a slower rate but evolve faster |
| (4) <i>Caenorhabditis elegans</i> | – A free-living pathogenic nematode used as vector in HGP |

- (1)
- (2)
- (3)
- (4)

122. State **True(T)** or **False(F)** to the given statements.

- Two nucleotides are linked through 3'-5' phosphodiester linkage to form a dinucleotide.
 - Lambda phage has single-stranded linear DNA with 48502 nucleotides.
 - Rho factor (ρ) is required for termination of transcription
 - UTRs are present at both 5'-end after start codon and 3'-end before stop codon.
 - M. Wilkins and Rosalind Franklin had proposed a definitive double helix model of DNA on the basis of X-ray diffraction pictures
- Choose the **correct** option.

a b c d e

- F T T T T
- T T F F T
- T F T F F
- T F F F T

- (1)
- (2)
- (3)
- (4)

123. The ground tissue is made up of all the following, **except**

- Phloem
 - Endodermis
 - Xylem
 - Pith
 - Medullary rays
 - Epidermis
- Select the **correct** option.

- f only
- b, d and e only
- b and f only
- a, c and f only

124. 'X' number of tRNAs are found for three stop codons.

In the light of above statement, **correctly** fill in the blank for 'X'.

- (1) X – One
- (2) X – Three
- (3) X – Zero
- (4) X – Two

125. Which scientist studied different types of animal cells and reported that cells had a thin outer layer known as the plasma membrane?

- (1) Matthias Schleiden
- (2) Theodore Schwann
- (3) Robert Hooke
- (4) Rudolf Virchow

126. Select the **correct** statement w.r.t. polysomes.

- (1) It represents the surface structure of bacteria that play a role in mating.
- (2) It is a form of inclusion body in which reserve material are stored.
- (3) These are formed when several ribosomes attach to a single mRNA
- (4) Ribosomes of the polysomes help to translate DNA into proteins

127. Read the following statements and select **True** (T) or **False** (F) for the given statements and choose the **correct** option.

- a. Integral membrane proteins, are always partially embedded in the membrane
- b. Peripheral proteins cannot be removed easily and their removal requires crude methods like using detergents
- c. Quasi fluid nature of lipids, enables lateral movement of protein within the overall bilayer
- d. Every chromosome essentially has a primary constriction on the sides of which disc shaped structures are present

a b c d

- (1) T F T T
- (2) T T F F
- (3) F F T T
- (4) T F F F

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

128. Which of the following is **not** included in the significance of meiosis?

- (1) Formation of gametes
- (2) Cell repair and growth
- (3) Crossing over
- (4) Introduction of variations

129. How many of the following phases involves cell organelle or cell structure duplication?

- a. Pre-mitotic phase
 - b. S-phase
 - c. Post-mitotic phase
 - d. Intrameiotic interphase
- Choose the **correct** option.

- (1) Two
- (2) Three
- (3) Four
- (4) One

130. If a cell in the G₁ phase has 10 chromosomes and 5 picograms of DNA, what would be the corresponding chromosome number and DNA content in the G₂ phase?

- (1) 10 chromosomes; 10 pg
- (2) 20 chromosomes; 5 pg
- (3) 5 chromosomes; 20 pg
- (4) 5 chromosomes; 10 pg

131. Select the option representing the **correct** set of hormones released by pars distalis and pars nervosa respectively.

- (1) ADH, ACTH
- (2) MSH, Oxytocin
- (3) Prolactin, Vasopressin
- (4) Oxytocin, Growth hormone

132. Choose the **correct** match w.r.t. diseases and their respective symptoms.

- (1) Ascariasis – Blockage of lymphatic vessels
- (2) Ringworms – Dry, scaly lesions on skin, nails and scalp
- (3) Filariasis – Blockage of intestinal passage and abdominal pain
- (4) Amoebiasis – Lips and finger nails turn gray to bluish in colour

133. The option which represents **incorrect** comparison between acquired and innate immunity, is that

- (1) Former is pathogen specific and latter is non-specific
- (2) Latter is present at the time of birth while former develops post birth after encounter with the pathogens
- (3) Former provides humoral immune response while latter provides cell mediated immune response
- (4) Latter includes various barriers that prevent entry of pathogen in body while former releases army of proteins called immunoglobulins in response to pathogen

134. A similarity between AIDS and cancer is that both

- (1) Are always caused due to viruses
- (2) Can spread by mere touch or physical contact with an infected person
- (3) Can be detected by technique based on the principle of antigen-antibody reactions
- (4) Can be treated by radiotherapy and immunotherapy

135.Assertion (A): Morphine which is extracted from latex of poppy plant is very useful in patients who have undergone surgery.

Reason (R): It is very effective sedative and pain killer. In the light of above statements, select the **correct** option.

- (1) Both (A) and (R) are false
- (2) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (3) (A) is true, (R) is false
- (4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

136. In human males, primary sex organ is situated within a pouch called 'X'. Select the **incorrect** option w.r.t. 'X'.

- (1) Testes are located outside the abdominal cavity within 'X'
- (2) 'X' is responsible for protecting the testes
- (3) 'X' provides suitable temperature necessary for spermatogenesis
- (4) 'X' helps in maintaining the temperature higher than the internal body temperature, necessary for proper functioning of the testes

137. The process of fusion of male and female gametes which leads to the formation of a diploid cell is called

- (1) Gametogenesis
- (2) Embryogenesis
- (3) Fertilization
- (4) Gestation

138. Select the **correct** statement w.r.t. healthy fertile humans.

- (1) Androgens are produced by Sertoli cells under the effect of LH
- (2) Leydig cells are found inside the seminiferous tubules of testis
- (3) Lack of menstruation may be the indicative of pregnancy in females
- (4) Development of blastocyst is followed by the formation of zygote

139. Plasmid is

- (1) A part of the chromosomal DNA of bacteria
- (2) Always linear and double-stranded
- (3) Always devoid of antibiotic-resistant genes
- (4) Capable of autonomous replication

140. Compared to a simple stirred-tank bioreactor, a sparged stirred-tank bioreactor exclusively contains a/an

- (1) Sampling port
- (2) Flat-blade impeller
- (3) Sparger
- (4) Oxygen delivery system

141. While performing PCR, an error occurred during the extension step what can be the possible consequence of this?

- (1) Single-stranded DNA templates will not be obtained from the source dsDNA
- (2) Primers will not attach to the 3' end of the template strand
- (3) Amplification of DNA would not occur
- (4) Desired DNA segment will still get amplified multifold

142. EcoRI recognises a specific palindromic nucleotide sequence where it cuts the nucleotide sequence between

- (1) Two bases at the centre of the palindromic sequence
- (2) Two bases that contain a single homocyclic ring with two nitrogen atoms
- (3) Two bases each with two heterocyclic rings
- (4) Two bases that are categorised under pyrimidines

143. The cranial meninx that is in close contact with the brain tissue is

- (1) Arachnoid
- (2) Pia mater
- (3) Dura mater
- (4) Sub-arachnoid

144. Read the following statements:

- (a) Myasthenia gravis is an autoimmune disorder characterised by rapid spasms in muscles
 - (b) Gout is a type of arthritis caused due to accumulation of uric acid crystals in joints.
 - (c) After menopause, females are more susceptible to osteoporosis as compared to fertile females.
- Choose the option with only **correct** statements.

- (1) a, b and c
- (2) Only a and b
- (3) Only b and c
- (4) Only c

145. Match column I with column II.

Column I	Column II
Coronary	
a. artery disease	(i) Congestion of lungs
b. Heart attack	(ii) Heart stops beating
c. Cardiac arrest	(iii) Heart muscle is suddenly damaged by an inadequate blood supply
d. Heart failure	(iv) Atherosclerosis

Select the **correct** option.

- (1) a(iv), b(iii), c(i), d(ii)
- (2) a(iv), b(i), c(ii), d(iii)
- (3) a(iv), b(ii), c(i), d(iii)
- (4) a(iv), b(iii), c(ii), d(i)

146. In a fully contracted skeletal muscle, length of which of the following remains the same in comparison to its resting stage?

- (1) 'I' band
- (2) 'H' zone
- (3) 'A' band
- (4) Sarcomere

147. Tetany can be caused/characterised by all of the following, **except**

- (1) Low Ca^{+2} level in body fluids
- (2) Wild contraction in muscles
- (3) Due to hyposecretion of thyrocalcitonin
- (4) Due to hyposecretion of parathyroid hormone

148. Nodal tissue of human heart which can generate the maximum number of action potentials is present in the

- (1) Right upper corner of the right atrium
- (2) Right upper corner of the left atrium
- (3) Lower left corner of the left atrium
- (4) Upper left corner of the left atrium

149. During blood coagulation in humans, which of the following is responsible for the conversion of prothrombin into thrombin?

- (1) Fibrin
- (2) Only Ca^{2+}
- (3) Thrombokinasase
- (4) Fibrinogen

150. Choose the **odd** one w.r.t. the location of gap junctions in humans.

- (1) Biceps present in arm
- (2) Striated involuntary muscle present in heart
- (3) Muscle present in wall of urinary bladder
- (4) In between adjacent muscle fibres present in the wall of intestine

151. Select the secondary metabolite among the following whose monomeric unit is glucose and does not possess any complex helices.

- (1) Inulin
- (2) Starch
- (3) Chitin
- (4) Cellulose

152. Match column I and column II w.r.t. hormones and their respective function.

Column I	Column II
a. CCK	(i) Inhibits gastric secretion and motility
b. Secretin	(ii) Stimulates secretion of H_2O and HCO_3^- from pancreas
c. GIP	(iii) Stimulates secretion of pancreatic enzymes and bile juice
d. Gastrin	(iv) Stimulates secretion of HCl and pepsinogen

Select the **correct** option.

- (1) a(iii), b(i), c(ii), d(iv)
- (2) a(i), b(iv), c(ii), d(iii)
- (3) a(iv), b(iii), c(i), d(ii)
- (4) a(iii), b(ii), c(i), d(iv)

153. Choose the **correct** statement among the following w.r.t. excretion in animals.

- (1) Terrestrial adaptation necessitated the production of most toxic nitrogenous waste like uric acid for water conservation.
- (2) Land snails and marine fishes mainly excrete uric acid and are called uricotelic animals.
- (3) Reptiles, birds and insects excrete nitrogenous wastes in the form of pellet or paste with maximum loss of water and are called ureotelic animals.
- (4) In ureotelic animals, some amount of urea may be retained in the kidney matrix to maintain a desired osmolarity.

154. Earth was supposed to have been formed about

- (1) 500 million years ago
- (2) 4.5 billion years back
- (3) 2000 million years ago
- (4) 450 million years ago

155. Match Column I and Column II w.r.t. humans.

Column I	Column II
a. Spermiogenesis	(i) Release of semen by the penis into vagina
b. Spermatogenesis	(ii) Changes in the sperm before fertilisation in female reproductive tract
c. Insemination	(iii) Transformation of spermatids into spermatozoa
d. Capacitation	(iv) Release of sperms into the seminiferous tubules

Choose the option with all **correct** matches.

- (1) a(i), b(ii), c(iii), d(iv)
- (2) a(iv), b(iii), c(i), d(ii)
- (3) a(iv), b(ii), c(iii), d(i)
- (4) a(iii), b(iv), c(ii), d(ii)

156. T.W. Engelmann worked on green alga, *Cladophora* along with aerobic bacteria and

- (1) Revealed the essential role of light in growth of green plants
- (2) Hypothesised that plants restore the air whatever breathing animals remove
- (3) Observed that the bacteria accumulated mainly in the region of blue and red light of the split spectrum
- (4) Showed that sun light is essential to the plant processes

157. Which of the following is **correct** match w.r.t. colour of photosynthetic pigments in chromatogram?

- (1) Xanthophylls – Yellow
- (2) Carotenoids – Green, Yellow
- (3) Chlorophyll *b* – Bright or blue-green
- (4) Chlorophyll *a* – Yellow-green

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

158. Both alcoholic and lactic acid fermentation are similar in all of the following features, **except**

- (1) Occurrence of decarboxylation
- (2) Hazardous end product are produced
- (3) Net gain is 2 ATP
- (4) Less than 7% of the energy in glucose is released

159. If 20,000 J of energy is trapped at producer level then calculate the energy transferred to top carnivore level in the given food chain?

Plant → Grasshopper → Frog → Snake

- (1) 0.2 kJ
- (2) 0.2 J
- (3) 200 kJ
- (4) 20 J

160. The common enzyme in Krebs' cycle and mitochondrial electron transport system is

- (1) NADH dehydrogenase
- (2) Pyruvate kinase
- (3) Succinate dehydrogenase
- (4) Phosphofructokinase

161. Which of the following is the **correct** expression to evaluate the net primary productivity?

- (1) $NPP = GPP \times R$
- (2) $NPP = GPP - R$
- (3) $NPP = GPP/R$
- (4) $GPP = NPP - R$

162. Which one of the following plants possess coralloid roots?

- (1) *Dryopteris*
- (2) *Pinus*
- (3) *Cycas*
- (4) *Chara*

163. Read the following features of the members related to a particular class of algae

- a. Major pigments are chlorophyll *a* and chlorophyll *b*
 - b. Food is stored in the form of starch
 - c. Cell wall is composed of cellulose
 - d. Found in freshwater, brackish water or salt water
- In the light of above features select the organism that belongs to the given group

- (1) *Ectocarpus*
- (2) *Polysiphonia*
- (3) *Chlamydomonas*
- (4) *Fucus*

164. Identify the **incorrect** statement w.r.t. mosses.

- (1) The members reproduce vegetatively by fragmentation.
- (2) The dominant phase of life cycle consists of foot, seta and capsule.
- (3) They have an elaborate mechanism of spore dispersal
- (4) The second gametophytic stage develops from the secondary protonema as a lateral bud

165. Select the **incorrectly** matched pair.

- (1) Tobacco Mosaic Virus – Compact rod-shaped virus
- (2) Adenovirus – Causes respiratory infections
- (3) LAB – Heterotrophic bacteria
- (4) *Saccharomyces* – Used in fermentation of Dosa and cheese

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

166. Swiss cheese is a product of fermentation carried out by

- (1) *Penicillium roquefortii*
- (2) *Penicillium camembertii*
- (3) *Propionibacterium sharmanii*
- (4) *Trichoderma polysporum*

167. Competition cannot occur

- (1) Between unrelated species
- (2) When resources are not limiting
- (3) Between closely related species
- (4) Between two unrelated species living in two different zones

168. Read the following statements and select the **correct** option.

Statement A: Tropical Amazonian rain forest in South America has the greatest biodiversity on earth.

Statement B: As we move from low to high latitude, biodiversity increases.

- (1) Both statements A and B are correct
- (2) Both statements A and B are incorrect
- (3) Only statement A is correct
- (4) Only statement B is correct

169. According to rivet popper hypothesis used by Paul Ehrlich, species is represented by

- (1) Airplane
- (2) Rivets
- (3) Wings
- (4) Seats

170. Mark the organism that breeds only once in its life time.

- (1) Pacific salmon fish
- (2) Most birds
- (3) Earthworm
- (4) Mango

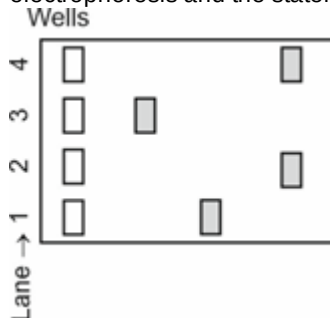
171. Choose the **correct** match w.r.t. male cockroaches.

- (1) Spermathecae – Present in the 6th abdominal segment
- (2) Testes – Lying laterally in the 2nd-6th abdominal segments
- (3) Anal style – Found attached to the 10th abdominal segment
- (4) Mushroom-shaped gland – Present in the 6th-7th abdominal segments

172. Choose the **incorrect** option w.r.t. the structure that is a component of brain stem but not the hind brain.

- (1) A canal named cerebral aqueduct passes through it
- (2) Located between the thalamus /hypothalamus of the forebrain and pons of the hindbrain
- (3) Receives and integrates visual, tactile and auditory inputs
- (4) Its ventral portion consists of four round swellings

173. Consider the given gel image obtained after gel electrophoresis and the statements w.r.t. it.



- a. The DNA fragment present in 'lane 1' is bigger in size than the DNA fragment present in 'lane 4'
 - b. The DNA fragments present in 'lane 2' and 'Lane 4' are possibly equal in size
 - c. The DNA fragment in 'lane 3' seems to be smaller in size than the DNA fragment present in 'lane-1' as it has moved closer to cathode due to the sieving effect of the gel
 - d. The wells are made towards the positive electrode in which bromophenol blue is also added that acts as the staining dye
- Choose the option with only **correct** statements.

- (1) a and d
- (2) b and c
- (3) a and b
- (4) d and c

174. Select the **odd** one w.r.t. *Obelia*.

- (1) Exhibits alternation of generation of body forms
- (2) Possesses a porous body
- (3) Reproduces sexually by formation of gametes
- (4) Exhibits extracellular as well as intracellular digestion

175. Choose the **correct** match.

- (1) *Pleurobrachia* – Presence of true coelom
- (2) *Ancylostoma* – Acoelomate
- (3) *Pila* – Exhibits internal fertilisation
- (4) *Euspongia* – Exhibits external fertilisation

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

176.Probe in molecular biology

- (1) Constitutes a double stranded RNA
- (2) Is used only for detection of foreign antibodies
- (3) Possesses a radioactive molecule
- (4) Is used as a conventional and old method of diagnosis

177.Select the **correct** option w.r.t., Rosie.

- (1) Transgenic animal produced in the year 1990
- (2) Transgenic animal which produced human protein enriched milk (2.4 gm per litre)
- (3) Milk produced by it was used to treat respiratory disease caused due to chronic smoking
- (4) Was made transgenically to produce milk in the same year in which gene therapy for the first time was given to a 4-year old girl

178.In which of the following ARTs, blastocyst with 32-celled stage is transferred into the uterus of the female who cannot conceive naturally?

- (1) GIFT
- (2) ZIFT
- (3) IUT
- (4) IUI

179.All of the following sexually transmitted diseases are completely curable, if detected early and treated properly, **except**

- (1) Syphilis
- (2) Genital warts
- (3) Genital herpes
- (4) Trichomoniasis

180.Which of the following factors favours the formation of oxyhaemoglobin?

- (1) Low pO_2
- (2) High temperature
- (3) High H^+ concentration
- (4) High pH